HUY HOANG CHU

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EDUCATION

Nanyang Technological University Singapore (NTU), Singapore.
PhD, Electrical and Electronics Engineering; Concentration; Nanophotonics, August 2029.
Thesis: "Superradiance Enhancement Via Photonics Metasurfaces."
GPA: N/A

Korea Advanced Institute of Science and Technology (KAIST), **Daejeon, South Korea.**
McS, Mechanical Engineering; Concentration; Applied Optics, August 2025.
Thesis: "Femtosecond laser-induced third harmonic generation (THG) for nonlinear optical characterization of semiconductor materials."
GPA: 3.47/4.3 (91.7/100)

Ho Chi Minh City University of Technology and Engineering (HCMUTE), Vietnam.
BE, Machine Manufacturing Technology; Concentration; Precision Engineering, November 2022
GPA: 8.76/10 – Valedictorian.

RESEARCH EXPERIENCE

CNRS-International-NTU-Thales Research Alliance (CINTRA), Singapore
Graduate Student Researcher, (August/2023 – August/2025)

- Superradiance Enhancement Via Photonics Metasurfaces.

Ultrafast Optics for Ultraprecision Technology Research Center of KAIST, South Korea
Graduate Student Researcher, (August/2023 – August/2025)

- Applied nonlinear optics for surface characteristics of semiconductor materials.
- Designed ultrathin metal-dielectric nanophotonic cavity for biosensing.
- Utilized frequency-comb-referenced multi-wavelength interferometry for high-speed 3D reconstruction of stepped micro-bumps in heterogeneous semiconductor packaging.
- Compensated the coherence envelope shifted error due to nonlinear phase change caused by dispersive properties of dissimilar material and thin film effect in optical measurements method.

HONORS & AWARDS

NTU Research Scholarship, Nanyang Technological University Singapore, 2025
Best Paper Award, ISMTTI 2025, 2025
KAIST Scholarship, Korea Advanced Institute of Science and Technology, 2023
SBA Scholarship, Saigon High-tech Park Business Association, Intel Vietnam, Jabil Vietnam, 2022
Graduated Valedictorian, Ho Chi Minh City University of Technology and Engineering (HCMUTE), 2022

PAPERS & PUBLICATIONS

Jun Hyung Park, Dae Hee Kim, **Huy Hoang Chu**, Ji Won Hahm, Guseon Kang, Dongil Lee, Seyong Song, Mingu Kang, Seung-Woo Kim and Young-Jin Kim, Frequency-comb-referenced multi-wavelength interferometry for high-speed 3D reconstruction of micro-bumps in heterogeneous semiconductor packaging, **Nanophotonics** 2024-0578, 2025.

Huy Hoang Chu, Dae Hee Kim, Jun Hyung Park, Sukkyung Kang, Jaiho Son, Hyunmin Lee, Hongki Yoo, Seung-Woo Kim, Sanha Kim, Young-Jin Kim, Nanometer-resolution White-light Scanning Interferometry for Surface-profiling of Hybrid Bonding Samples for Advanced Semiconductor Packaging, **Applied Surface Science**, vol. 689, 162466, 2025.

PRESENTATIONS

Huy Hoang Chu. (2025). *Femtosecond Laser-Induced Third Harmonic Generation (THG) for Nonlinear Optical Characterization of Semiconductor Materials*, [Oral presentation] **ICMDT 2025** (Accepted)**,** Himeji, Japan.

Huy Hoang Chu. (2025). *Nonlinear Optical Characterization of Semiconductor Wafers via Femtosecond Laser-Induced Third Harmonic Generation (THG)*, [Oral presentation] **ISMTTI 2025** (Accepted), Wuhan, China.

PROFESSIONAL EXPERIENCE

Samsung HCMC Research and Development Center, HCMC, Vietnam
R&D Engineer, W/M R&D Lab, (August/2022 – August/2023)

- New Product Development (NPD) for washing machines from initiation design to release mass production.

SERVICE & LEADERSHIP

Mechanical Engineering International Student Organization of KAIST (MEISO), South Korea.
Student Leader, (May/2024 – August/2025)

- Hosted various events for international students in my department, including welcoming freshmen, orientation sessions for new students, New Year and end-of-year gatherings, and several seminars covering career opportunities in Korea for international students, graduate procedures, and settling into life at KAIST
- Supported Vietnamese and International students interested in joining the KAIST ME department.

SKILLS & INTERESTS

- Research Interests: Nanophotonic, Optical Metrology, Interferometry.
- Skills: Experimental Optics, Material characterization (AFM, WSI, CLSM), Nanofabrication (FIB).
- Technical Proficiencies: Ansys Lumerical, MATLAB, Lightwave Explorer, COMSOL
- Languages: Vietnamese: Native, English: IELTS Academic 7.0 Overall (2022), Korean: Basic.